



# Physiology of the Large Intestine

Physiology GIT Block · MEDucation by Mattin Majid

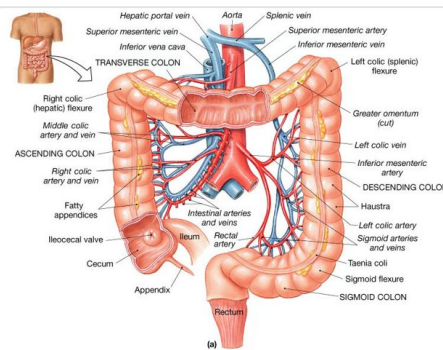
## BEFORE YOU READ — ANSWER FROM MEMORY

1. Name at least four functions of the large intestine.
2. What are the three characteristic movements of the colon, and how do they differ?
3. Which portion of the colon does most of the absorbing, and which mainly stores feces?

*Guessing wrong before reading still improves retention — attempt all three, then check as you go.*

## 1 Regions of the Colon

Cecum with appendix → ascending colon → transverse colon → descending colon → sigmoid colon → rectum → anal canal.



Colon regions and their vascular supply.

## 2 Functions of the Large Intestine

| # | Function                                                                                                                                                                                           |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Haustral churning, peristalsis, and mass peristalsis drive contents of colon into rectum                                                                                                           |
| 2 | Bacteria convert proteins to amino acids and produce B vitamins — <b>thiamine (B1)</b> , <b>B12</b> , <b>folic acid</b> — and <b>vitamin K</b> , contributing to erythropoiesis and blood clotting |
| 3 | Absorbs water, ions, vitamins, organic substances (e.g. glucose), and alcohol/drugs (anaesthetics, sedatives, steroids)                                                                            |
| 4 | Formation of feces                                                                                                                                                                                 |
| 5 | Defecation (emptying rectum)                                                                                                                                                                       |
| 6 | Excretes heavy metals — mercury, lead, bismuth, arsenic — via feces                                                                                                                                |
| 7 | Secretes mucin and inorganic substances (chlorides, bicarbonates)                                                                                                                                  |

## 3 Secretions — mucus only

**NEVER MISS:** the large intestine has **no villi** and its epithelial cells contain **almost no enzymes**. Crypts of Lieberkühn secrete mucus — that's it. All further chemical digestion here is bacterial, not enzymatic.

Mucus lubricates the colon and protects the mucosa. Pelvic nerve (parasympathetic) stimulation increases both mucus secretion and peristaltic motility. Under **extreme** parasympathetic stimulation, a person can pass ropy mucus as often as every 30 minutes — often with little or no fecal material.

## 4 Motility — three distinct movements

| Movement                 | Mechanism                                                                                                                        | Rate             | Trigger          |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------|------------------|
| <b>Haustral churning</b> | Haustra distend as they fill; once distension reaches threshold, the wall contracts and squeezes contents into the next haustrum | Continuous, slow | Local distension |

|                         |                                                                                 |                                                       |                                                       |
|-------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|
| <b>Peristalsis</b>      | Standard forward wave                                                           | 3–12 contractions/min — slower than more proximal gut | Ongoing                                               |
| <b>Mass peristalsis</b> | Strong wave from mid-transverse colon, quickly driving contents into the rectum | 3–4×/day                                              | <b>Gastrocolic reflex</b> — food entering the stomach |

## 5 Ileocecal valve regulation

|                         |                                                                                                                                  |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>At rest</b>          | Normally partially closed — chyme passes into the cecum slowly                                                                   |
| <b>After a meal</b>     | <b>Gastroileal reflex</b> intensifies ileal peristalsis, forcing chyme into the cecum; <b>gastrin</b> also relaxes the sphincter |
| <b>Cecal distension</b> | Intensifies contraction of the ileocecal sphincter — a local negative feedback loop                                              |

## 6 Chemical digestion — bacteria do the work

|                      |                                                                                                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Carbohydrates</b> | Bacteria ferment remaining carbohydrate → H <sub>2</sub> , CO <sub>2</sub> , methane gases → flatus (flatulence if excessive)                                                                                                                          |
| <b>Proteins</b>      | Bacteria convert remaining protein to amino acids, then break these down into indole, skatole, H <sub>2</sub> S, fatty acids. Some indole/skatole is eliminated in feces (odor); the rest is absorbed, detoxified by the liver, and excreted in urine. |
| <b>Bilirubin</b>     | Bacteria decompose bilirubin into simpler pigments including <b>stercobilin</b> — this is what gives feces their brown color                                                                                                                           |

## 7 Absorption & feces formation

| Region                        | Role                                                    |
|-------------------------------|---------------------------------------------------------|
| <b>Proximal half of colon</b> | <b>"Absorbing colon"</b> — most absorption happens here |
| <b>Distal colon</b>           | <b>"Storage colon"</b> — holds feces until defecation   |

After 3–10 hours in the large intestine, chyme becomes solid/semisolid feces via water absorption. Feces = water, inorganic salts, sloughed epithelial cells, bacteria, bacterial decomposition products, unabsorbed digested material, and indigestible food parts. Of the 0.5–1.0 L of water entering the colon daily, only ~100–200 mL is **not** absorbed (the rest is absorbed via osmosis) — a small addition on top of the ~90% already absorbed in the small intestine, but still important for water balance.

| Reabsorbed in the large intestine               |
|-------------------------------------------------|
| Water                                           |
| Vitamins — K, biotin (B7), B5                   |
| Organic wastes — urobilinogens, sterobilinogens |
| Bile salts                                      |
| Toxins                                          |

## 8 The Rectum & continence

|                            |                                                                                                                                                                                                      |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Rectum</b>              | Last portion of the digestive tract; terminates at the anal canal; has internal and external anal sphincters                                                                                         |
| <b>Sensory innervation</b> | Mechanoreceptors in the rectum detect distension and supply the ENS. The anal canal's skin region is innervated by somatosensory nerves to the CNS, with receptors for pain, temperature, and touch. |
| <b>Continence</b>          | Contraction of the internal anal sphincter and puborectalis muscle blocks the passage of feces and maintains continence                                                                              |

### P Must-remember pearls

- The large intestine secretes **mucus only** — no digestive enzymes. Digestion here is bacterial, not enzymatic.
- Mass peristalsis (3–4×/day, via the **gastrocolic reflex**) is what actually drives contents into the rectum — distinct from continuous haustral churning.
- Proximal colon = "absorbing colon"; distal colon = "storage colon."
- Only ~100–200 mL of the 0.5–1.0 L of water entering the colon each day is **not** absorbed.
- **Stercobilin**, from bacterial bilirubin breakdown, is what gives feces their brown color.

### Q Test yourself

**1. A patient passes ropy mucus every 30 minutes with little fecal material. This suggests:**

- A. Ileocecal sphincter failure
- B. Extreme parasympathetic stimulation of the colon
- C. Loss of haustral churning
- D. Bacterial overgrowth in the cecum only

**2. Which colon segment is responsible for most water and electrolyte absorption?**

- A. Proximal (ascending/transverse) colon
- B. Distal (descending/sigmoid) colon
- C. Rectum
- D. Anal canal

**3. The gastrocolic reflex is responsible for triggering:**

- A. Haustral churning
- B. Mass peristalsis
- C. Ileocecal sphincter relaxation only
- D. Mucus secretion

**4. A patient with pale, clay-colored stools most likely has reduced colonic delivery of:**

- A. Bile salts
- B. Bilirubin (reducing stercobilin formation)
- C. Mucus
- D. Short-chain fatty acids

**5. Which vitamins do colonic bacteria help produce?**

- A. Vitamin C and folate only
- B. Thiamine (B1), B12, folic acid, and vitamin K
- C. Vitamin A and D
- D. Vitamin E only

**ANSWERS — COVER UNTIL YOU'VE COMMITTED**

- 1 — B** — extreme parasympathetic drive can produce frequent ropy mucus discharge with minimal fecal content.
- 2 — A** — the proximal colon is the "absorbing colon"; the distal colon mainly stores.
- 3 — B** — food entering the stomach triggers the gastrocolic reflex, producing mass peristalsis 3–4×/day.
- 4 — B** — reduced bilirubin delivery (e.g. biliary obstruction) means less bacterial conversion to stercobilin, the brown pigment.
- 5 — B** — colonic bacteria produce thiamine, B12, folic acid, and vitamin K.

### SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam